

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

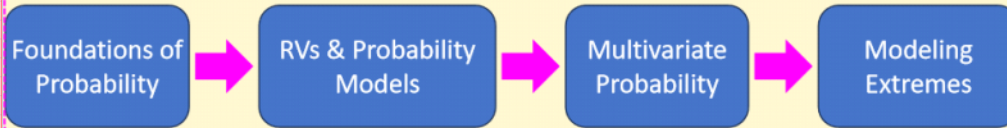
Dr. Prakash S Badal

Announcement

- Lectures
 - Quiz-2 on 12th October at 8 AM

Course Flow

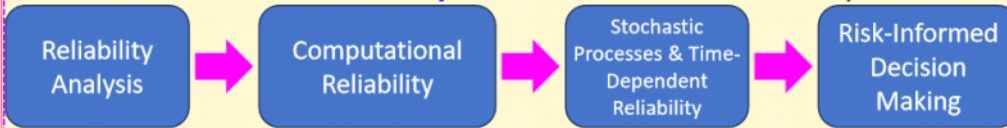
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data



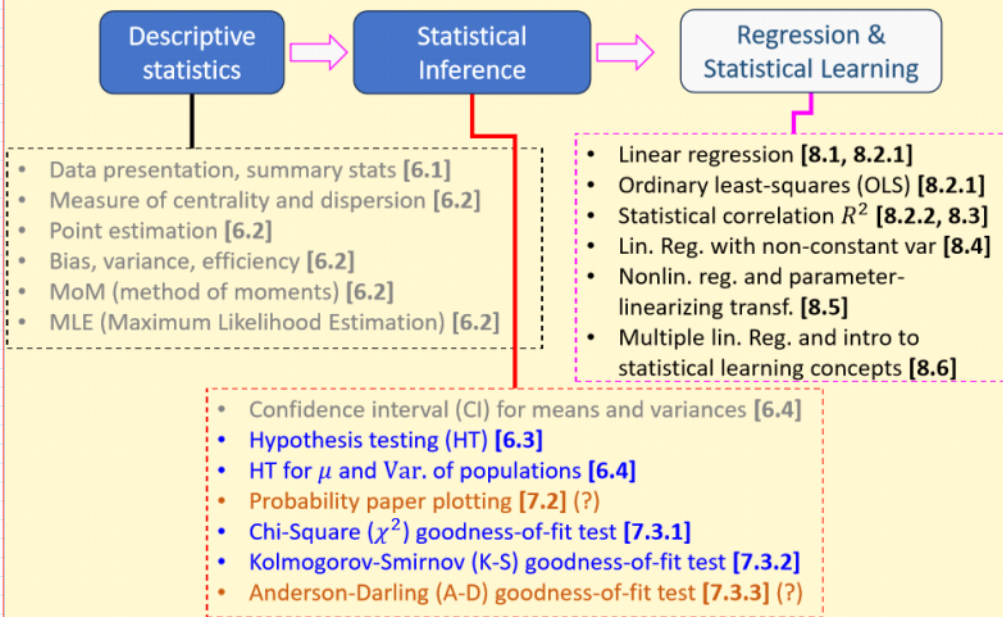
Module 3: **Reliability** – Decision under Uncertainty



Statistical Inference (M2C2)

Module II: Chapter 2 (M2C2)

Module 2: **Statistics** – Learning from Data



M2C2

Interval Estimation: two-sided and one-sided CI for population μ and σ^2 under known and unknown variance assumptions.

Hypothesis Testing Principles: H_0 and H_1 .

Error Analysis & Test Power: Type I (α) and Type II (β) error probabilities, p -values.

Parametric Testing: Z -tests, t -tests, χ^2 -tests, and F -tests for single-sample and two-sample population parameters.

Empirical Probability Plotting: probability paper and ECDFs to evaluate candidate dist. visually.

Goodness-of-Fit Verification: χ^2 , Kolmogorov-Smirnov (K-S), and Anderson-Darling (A-D) non-parametric hypothesis tests to validate distribution fits.

Motivation

How to test hypotheses and validate probability models?

Hypothesis Testing

Error Analysis & Test Power

Flow

- Null/alternative hypothesis
- Type I/II error
- Hypothesis testing method
- Error calculations
- Power of test
- p -value
- GoF

Central limit theorem: recap

- For large sample size n , **sample mean** approaches

a **normal distribution**,

with **mean μ** (same as population mean), and

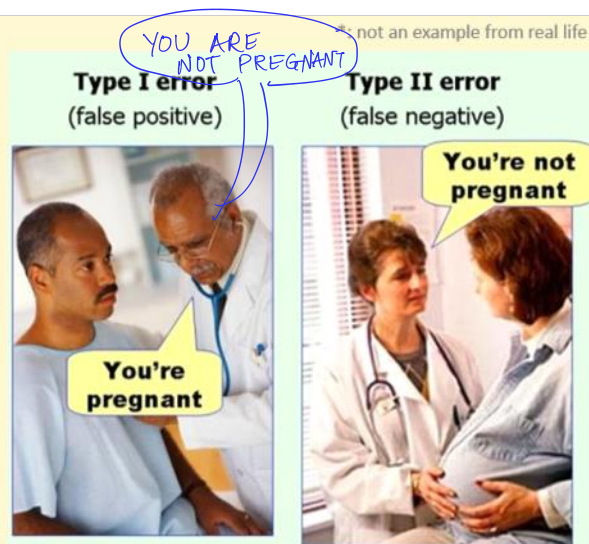
variance σ^2/n (less than population Var.).

Recall spreadsheet demo!

9

Spot the error*

Accuracy of test
False Positive
False Negative
Outcome of the test



H_0 : NULL HYPOTHESIS: YOU ARE NOT PREGNANT
DEFAULT

H_1 : ALTERNATIVE HYPOTHESIS: YOU ARE PREGNANT

10

Justice system in a democracy

- How can someone be arrested if they really are **presumed innocent**?
- Why is a defendant pronounced "**Not Guilty**" instead of innocent?
- Why do citizens put up with a system that allows **criminals to go free on technicalities**?

11

Common challenges in justice, research (life?)

- **Strong Urge** to believe in an Unusual/Rare event
 - defendant is guilty
 - New medicine cures cancer
 - New concrete stronger than existing.
- **Serious Repercussions of Making a Wrong decision**

Two kinds of errors are possible

YOU ACTED ON YOUR URGE, BUT IT TURNED OUT TO BE FALSE		YOU DID NOT ACT ON YOUR URGE, AND IT TURNED TO BE TRUE	
Innocent goes to Jail		Guilty goes free	
Patients getting the medicine.		Patients suffering/dying, even though the cure exists.	

- **No SURE SHOT WAY TO KNOW TRUTH.**
- **LIMITED DATA.**

12

Hypothesis testing is the solution!

13

Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

1. **The alternative hypothesis:** H_1 or H_a This is the reason that criminal is arrested.
2. **The null hypothesis:** H_0 The presumption of innocence
3. **A standard of justice:** Beyond reasonable doubt.
4. **A data sample:** Partial information.

Hypothesis testing (& justice system)

Both statistical testing and the justice system:

1. Concentrate on "Rejecting the Null Hypothesis."

- It is easier
- Rejection of "Presumption of Innocence"
≡ Pronounced Guilty

Hypothesis testing (& justice system)

Neither statistical testing and the justice system are perfect:

- Sometimes, the jury makes an error.
 - An innocent person goes to jail
 - Statisticians call it a Type I error
- Sometimes, a guilty person is set free
 - Statisticians call it a Type II error

WHICH ONE IS WORSE?

Citizens find Type II error disturbing, however it is NOT as horrific as Type I.

Type I error = $\left\{ \begin{array}{l} \text{An innocent person} \rightarrow \text{Jail} \\ + \\ \text{A Guilty walking free} \end{array} \right.$

Errors in Hypothesis testing (& justice system)

- An innocent person goes to jail *Type I error*
- A guilty person is set free *Type II error*

HT & Justice System put a lot of emphasis on Reducing Type I error.

Airport Security: Type II worse.

Errors in Hypothesis testing (& justice system)

HT & justice lady put a lot of emphasis on

avoiding Type I error.

H_0 : YOU ARE INNOCENT.
 H_1 : YOU ARE ^{THE} CRIMINAL.

Type I: Innocent $\rightarrow$ Jail
 $H_0 \checkmark$

H_0 is True $\rightarrow$ you acted as if H_0 is False

H_0 is False $\rightarrow$ you acted as if H_0 is True

Product example:

- Null hypothesis, H_0 : a product satisfies customer requirements.
- If H_0 is rejected, do not sell the product to customers.

*Type I error: $\nearrow$ Reject a Good Batch.
 $\rightarrow$ Huge economic loss.*

Type II error: Failing to reject a Bad Batch & shipping it to the customer $\rightarrow$ Reputation loss.

Which one is worse? *TYPE II worse*

Spot the error type

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO₂. A fuel company wants to test a new type of gasoline designed to have lower CO₂ emissions. Here are their hypotheses:

$$H_0 : \mu = 8.9 \text{ kg}$$

$$H_a : \mu < 8.9 \text{ kg}$$

(where μ is the mean amount of CO₂ emitted by burning one gallon of this new gasoline).

Under which of the following conditions would the company commit a Type I error?

H_0 ✓

Choose 1 answer:

- ☐ A The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ B The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ C The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ D The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.

A large university is curious if they should build another cafeteria. They plan to survey a sample of their students to see if there is strong evidence that the proportion interested in a meal plan is higher than 40%, in which case they will consider building a new cafeteria.

Let p represent the proportion of students interested in a meal plan. Here are the hypotheses they'll use:

$$H_0 : p \leq 0.40$$

$$H_a : p > 0.40$$

What would be the consequence of a Type II error in this context?

Choose 1 answer:

- ☐ A They don't consider building a new cafeteria when they should.
- ☐ B They don't consider building a new cafeteria when they shouldn't.
- ☐ C They consider building a new cafeteria when they shouldn't.
- ☐ D They consider building a new cafeteria when they should.

19

Spot the error type

A quality control expert wants to test the null hypothesis that a new solar panel is no more effective than the older model.

What would be the consequence of a Type I error in this context?

Choose 1 answer:

- ☐ A They don't conclude the new panel is more effective when it actually is not more effective.
- ☐ B They don't conclude the new panel is more effective when it actually is more effective.
- ☐ C They conclude the new panel is more effective when it actually is more effective.
- ☐ D They conclude the new panel is more effective when it actually is not more effective.

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO₂. A fuel company wants to test a new type of gasoline designed to have lower CO₂ emissions. Here are their hypotheses:

$$H_0 : \mu = 8.9 \text{ kg}$$

$$H_a : \mu < 8.9 \text{ kg}$$

(where μ is the mean amount of CO₂ emitted by burning one gallon of this new gasoline).

Which of the following would be a Type II error in this setting?

Choose 1 answer:

- ☐ A The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ B The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ C The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ D The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.

20

L-30 (2/2; HT)

Oct 05, 2026

Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

1. The alternative hypothesis:

This is why a *criminal is arrested* (or a research undertaken)
denoted by H_1 or H_A .

2. The null hypothesis: → default Notion

The *presumption of innocence* (the solution won't work!)
 H_0 is the logical opposite of H_1 .

3. A standard of justice:

Absolute proof IMPOSSIBLE, hence a *reasonable doubt* (sig. level)
Reject the null hypothesis beyond a reasonable doubt.

4. A data sample:

Evaluation of *partial information* (test score)
The "whole truth and nothing but the truth" is often impossible.

Errors in Hypothesis testing (& justice system)

$$\text{Type I error} \equiv \left\{ \begin{array}{c} \text{an innocent person goes to jail} \\ + \\ \text{a guilty person walks free} \end{array} \right\}$$

HT (& justice system) puts a lot of emphasis on avoiding Type I error.

Counter example: Airport security

Innocent person and alarm goes off (Type I error)

Guilty person and alarm goes not go off (Type II error)

Sometimes, Type II error is worse!

e.g., airport security.

When the consequence of Type II error far outweighs that of type I

Example 1

- Consider bars made of IS 1786-compliant **Fe500** grade steel $f_y = 500 \text{ MPa}$
 - Let's focus on the mean yield strength
 - We are interested in deciding whether or not the mean yield strength is 500 MPa

Formally, $H_0: \mu_{F_y} = 500 \text{ MPa}$
 $H_1: \mu_{F_y} \neq 500 \text{ MPa}$

Four situations may arise

Decision ↓	H_0 is True	H_0 is False
Fail to Reject H_0	NOT an error	Type II error (β)
Reject H_0	Type I error (α)	NOT an error

Example 1

$$H_0: \mu_{F_y} = 500 \text{ MPa}$$

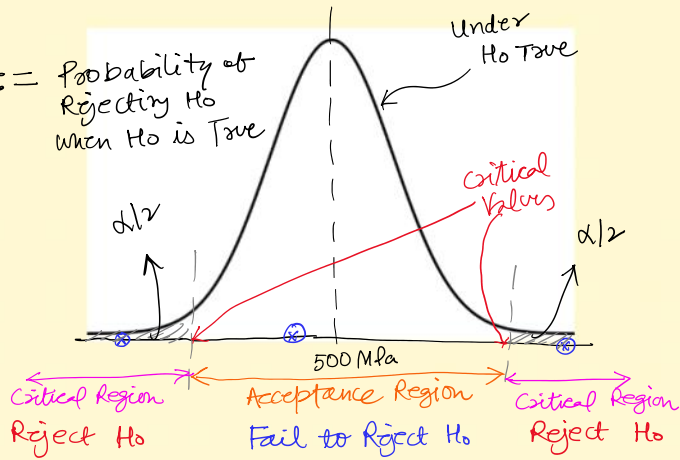
$$H_1: \mu_{F_y} \neq 500 \text{ MPa}$$

1. Write down the two hypotheses

2. Draw the PDF

3. Mark the critical value

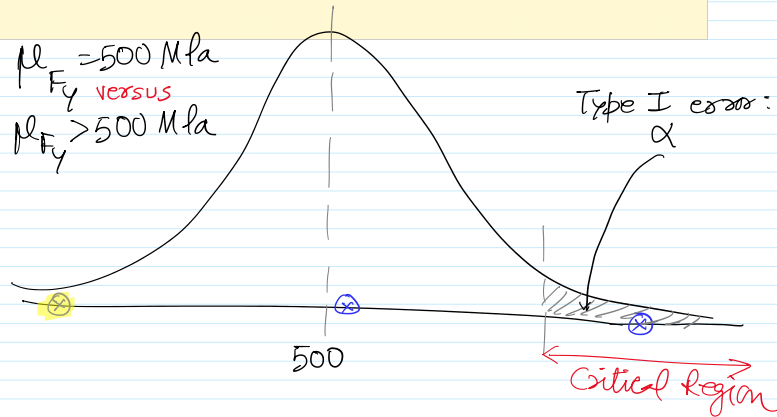
α := Probability of Rejecting H_0 when H_0 is True



$$H_0: \mu_{F_y} = 500 \text{ MPa}$$

versus

$$H_1: \mu_{F_y} > 500 \text{ MPa}$$



Type I error

26

Probability of Type I Error, α

- $\alpha = \text{Pr}(\text{Type I error})$
= $\text{Pr}(\text{Reject } H_0 \text{ when } H_0 \text{ is True})$
aka "the significance level"
or α -error

HOW TO REDUCE α -error?

- * widening the acceptance Region
- * Increase the sample size

Probability of Type I Error, α

- Consider Fe500 bar again; suppose $\sigma_{Fe500} = 25 \text{ MPa}$
- Suppose we tested a sample with $n = 10$ specimens
- If a client accepts bars with average $f_y \in [485 \text{ MPa}, 515 \text{ MPa}]$
- Find the **probability of Type I error**?

$$H_0: \mu_{F_y} = 500 \text{ MPa}$$

$$H_1: \mu_{F_y} \neq 500 \text{ MPa}$$

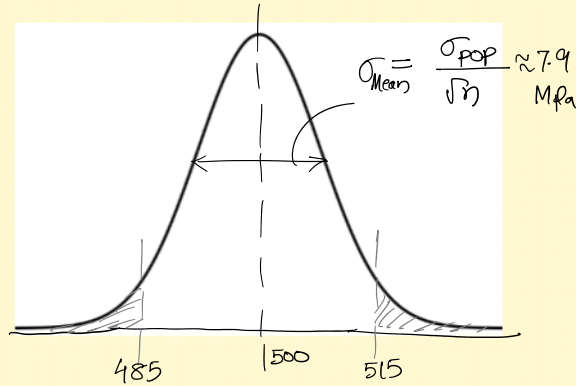
$\alpha = \Pr(\text{Rejecting } H_0 \text{ when } H_0 \text{ is True})$

$$= \Pr(\bar{X} < 485 \mid \mu = 500, \sigma = 7.9)$$

$$+ \Pr(\bar{X} > 515 \mid \mu = 500, \sigma = 7.9)$$

$$= \Pr(Z < -1.90) + \Pr(Z > 1.90)$$

$$\Rightarrow \alpha = 2 \times 0.0287 = 5.74\%$$



Standard Normal Probabilities

Z-table

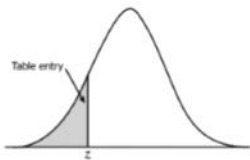


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Type II error

30

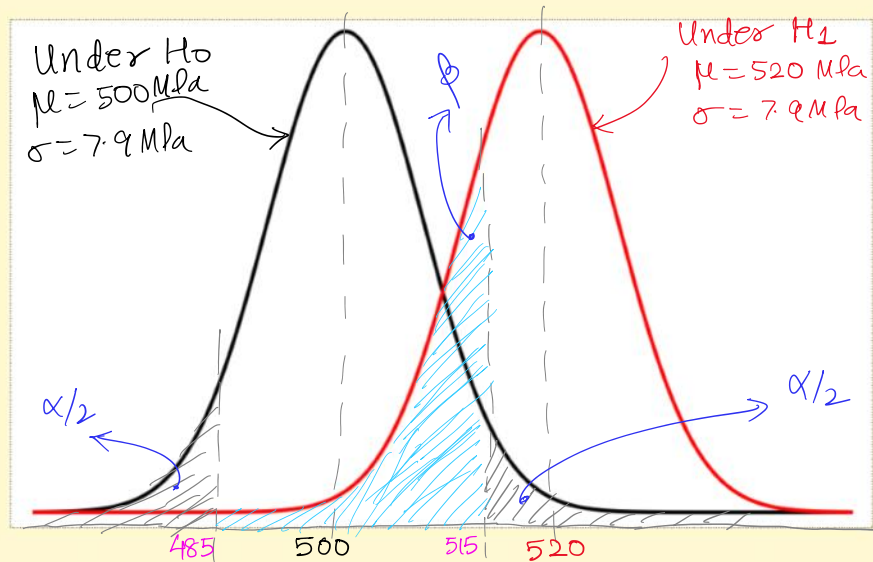
Probability of Type II Error, β

- $\beta = \text{Pr}(\text{Type II error})$
= $\text{Pr}(\text{failure to Reject } H_0 \text{ when } H_0 \text{ is False})$
aka the β -error

To calculate β -error, H_1 MUST be specific.

e.g. $H_0:$
 $\mu_{F_y} = 500 \text{ Mpa}$ versus $H_1:$
 $\mu_{F_y} = 520 \text{ Mpa}$

Type II error



Probability of Type II Error, β

- Fe500; $\sigma_{Fe500} = 25 \text{ MPa}$; we tested a sample with $n = 10$ specimens
- If an alternative hypothesis exists that $\mu_{F_y} = 520 \text{ MPa}$
- Find the **probability of Type II error**?

$$\beta = \Pr(\text{failure to Reject } H_0, \text{ when } H_0 \text{ is False})$$

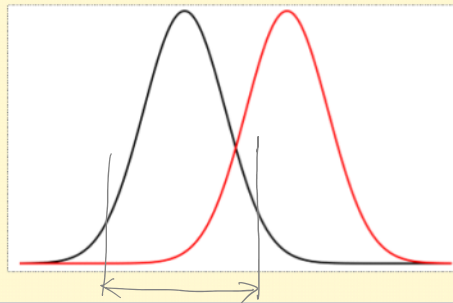
$$= \Pr(\bar{X} \in [485, 515] \mid \mu = 520 \text{ MPa}, \sigma = 7.9 \text{ MPa})$$

$$= \Pr\left(Z < \frac{515 - 520}{7.9}\right) - \Pr\left(Z < \frac{485 - 520}{7.9}\right)$$

$$= \Phi(-0.63) - \Phi(-4.42)$$

$$\approx 0.264 - 0.000$$

$$\approx 0.264$$



Power of a statistical test

- The **power** of a statistical test

$$\text{Power of Test} = 1 - \beta$$

Probability of correctly Rejecting H_0 ,
e.g., sending guilty to jail.

decision ↓	H_0 is True	H_0 is False
Fail to reject H_0	NOT an error	Type II error (β)
Reject H_0	Type I error (α)	$1 - \beta \rightarrow \text{Power}$ (NOT an error)

we control α 

Power: steel strength example

- In the steel strength problem, we found that $\beta = 0.2643$ when the true mean is $\mu = 520$, what is the power of this test

$$\text{Power} = 1 - \beta = 0.7357$$

Test is 73.6% powerful.

p-value

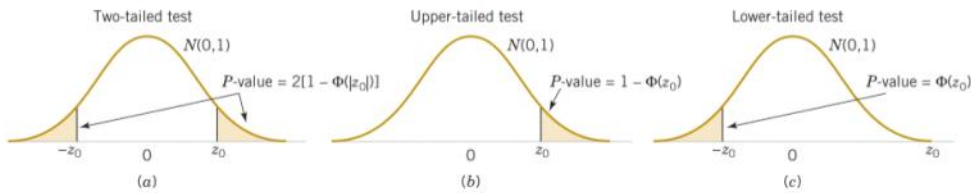


FIGURE 9-10 The P -value for a z -test. (a) The two-sided alternative $H_1: \mu \neq \mu_0$. (b) The one-sided alternative $H_1: \mu > \mu_0$. (c) The one-sided alternative $H_1: \mu < \mu_0$.

p-value

If there is truly no effect
how likely it is that Randomness alone
would give a sample as Extreme.

36

General Procedure for HT

- Once the question of interest is decided, and the experiment is designed:
 - How the data will be collected
 - How many observations are required
- The following steps are required
 1. Spell out H_0 and H_1
 2. Specify the test statistic to be used (such as Z_0)
 3. Specify the location of the critical region (two-tailed, upper-tailed or lower-tailed)
 4. Specify the criteria for rejection (typically, the value of α)
 5. Reject H_0 if $z_0 > z_{\alpha/2}$
 6. Calculate the power of the test, if H_1 is specific

37

Parametric Testing

- z -test; t -test
- χ^2 -test for goodness-of-fit test (7.3.1)
- F -tests for single-sample & two-sample pop. params.
- Kolmogorov-Smirnov (K-S) goodness-of-fit test

χ^2 -test for Goodness of Fit

Ang and Tang (Section 7.3.1)

Thank you!